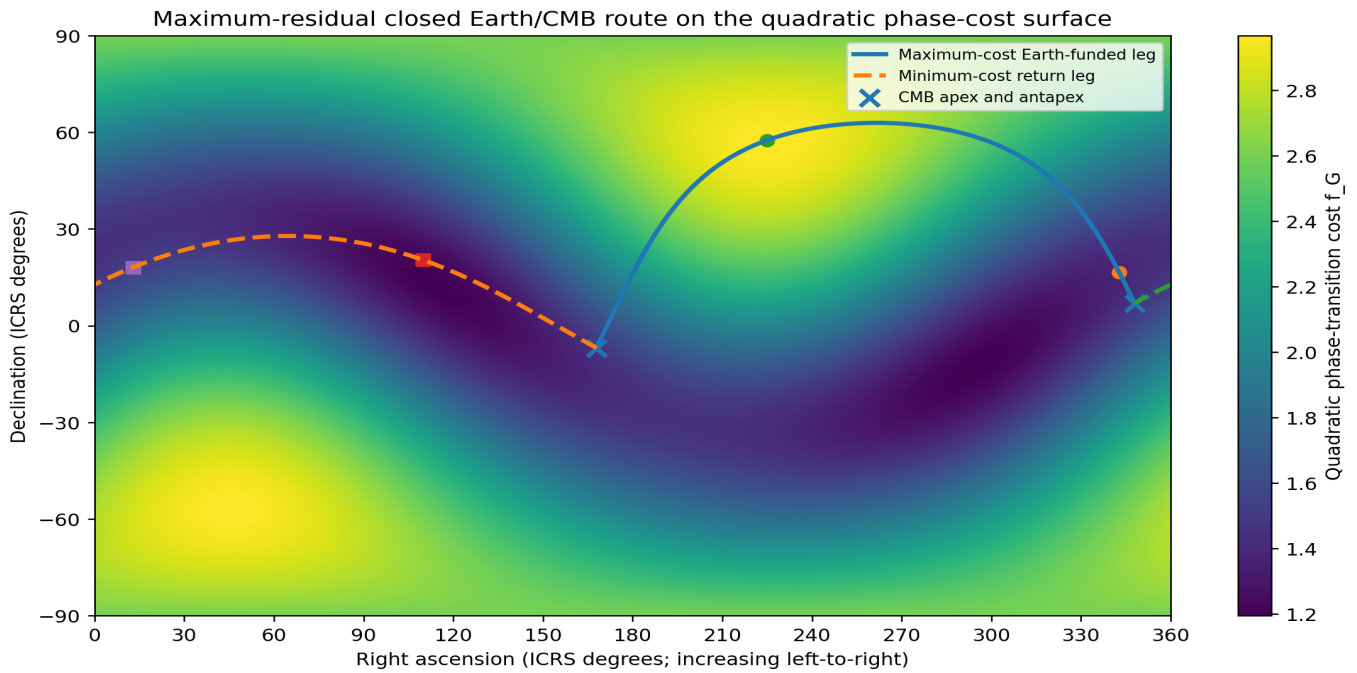


K-FIELD

Maximum Residual Closed CMB Route

Global integrated-cost optimization of the Earth-funded leg, minimum-cost return, closed spherical lune, residual function, and native Cell3 geometry



Title figure. The globally maximum integrated-cost Earth-funded CMB meridian and globally minimum integrated-cost return meridian form an exactly closed two-great-circle loop on the authoritative rectangular ICRS phase-transition cost surface.

Principal Investigator

P. Dwayne Esterline

BS, Huntington University (1999)

MBA, Indiana Wesleyan University (2008)

Krytronx

PROPRIETARY CONFIDENTIAL TRADE SECRET

Archival technical reference. Distribution and use are restricted. All numerical figures are generated from the exact matrix, vector, path integrations, route optimization, and native Cell3 geometry stated in this report.

Publication epoch: 1784650136

Contents

Contents	2
Abstract	3
1. Coordinate frame, state space, and assumptions	3
2. Quadratic phase-transition cost	3
3. Earth-forced meridian trajectories	4
4. Integrated path cost and closed-cycle objective	4
5. Numerical optimization method	5
6. Globally maximum-cost Earth-funded leg	6
7. Globally minimum-cost return leg	6
8. Closed spherical lune geometry	7
9. Maximum residual and residual function	8
10. Why this path differs from modulation extrema	9
11. Earth-force work and the required coupling asymmetry	9
12. Physical interpretation of the optimized cycle	9
13. Minimum mechanism requirements and testable consequences	9
14. Native Cell3 base geometry	10
14.1 Complete centered vertex table	12
14.2 Complete centered half-step node table	12
15. Reproduction procedure	13
16. Canonical result statement	13
Archival notice	13

Abstract

This paper identifies the actual single-pass closed path that maximizes residual integrated phase cost under the Earth/CMB forcing geometry. The directional state lies on the authoritative ICRS unit sphere. Earth forcing generates a meridional foliation: every admissible forced trajectory is a great-circle semicircle joining the CMB antapex and apex. Distinct forced meridians intersect only at those two endpoints. Therefore, the only nontrivial closed loop assembled from two distinct Earth-forced integral curves is a two-meridian spherical lune.

The cycle objective is defined by the integrated phase-transition cost $C[\gamma] = \int f_G ds$. The Earth-funded leg is selected to maximize this integral from antapex to apex. The return leg is selected independently to minimize the same integral from apex to antapex. Because both legs have fixed common endpoints and because the objective is additive, the global closed-cycle optimum separates exactly into the maximum meridian cost minus the minimum meridian cost.

The maximum Earth-funded leg has integrated cost 7.053342663751261. The minimum-cost return has integrated cost 4.310950793504075. Their difference is 2.742391870247186, equal to 38.8807406783 percent of the funded-leg path cost before switching and dissipative losses. The optimized planes are nearly orthogonal and enclose a spherical lune of 3.133544940435163 steradians, approximately 24.9359583335 percent of the sphere.

The report also distinguishes integrated-cost optimization from peak-to-peak modulation optimization, derives the route-independent work of the bare tangent-projection Earth force, identifies the required coupling asymmetry for residual retention, and records the full native Cell3 base geometry with all eight vertices, all 27 centered half-step nodes, and all primitive face planes.

1. Coordinate frame, state space, and assumptions

The directional state is a unit vector n in the fixed International Celestial Reference System Cartesian basis. The x axis is RA 0 deg and Dec 0 deg, the y axis is RA 90 deg and Dec 0 deg, and the z axis is Dec +90 deg. Right ascension increases from 0 to 360 deg without inversion, reflection, or conversion to another frame.

$$S^2_{\text{ICRS}} = \{ n \in \mathbb{R}^3 : n^T n = 1 \}$$

$$n(\text{RA}, \text{Dec}) = \begin{bmatrix} \cos(\text{Dec}) \cos(\text{RA}), \\ \cos(\text{Dec}) \sin(\text{RA}), \\ \sin(\text{Dec}) \end{bmatrix}^T$$

The device is placed at the Earth center. This removes surface latitude, longitude, and daily body-rotation geometry. Seasonal modulation of the Earth velocity is neglected. The Earth/CMB unit vector is therefore fixed in ICRS throughout the cycle.

$$\hat{c} = \begin{pmatrix} -0.970770855835935, \\ 0.207348009110986, \\ -0.120874929481992 \end{pmatrix}$$

$$\begin{aligned} \text{CMB apex:} \quad & \text{RA} = 167.943299879828 \text{ deg}, \text{ Dec} = -6.942599922130 \text{ deg} \\ \text{CMB antapex:} \quad & \text{RA} = 347.943299879828 \text{ deg}, \text{ Dec} = +6.942599922130 \text{ deg} \end{aligned}$$

2. Quadratic phase-transition cost

The directional phase cost is generated by the symmetric positive-definite matrix G_{phase} in the same authoritative ICRS basis. No eigenframe rotation is used for the route computation or for the sky maps.

$$G_{\text{phase}} = \begin{bmatrix} 2.9764106378 & 1.2431507795 & -2.0803600354 \\ 1.2431507795 & 2.5922286635 & -2.3178839937 \\ -2.0803600354 & -2.3178839937 & 6.7735709133 \end{bmatrix}$$

$$\begin{aligned} q(n) &= n^T G_{\text{phase}} n \\ f_G(n) &= \sqrt{q(n)} \end{aligned}$$

The local cost f_G is dimensionless in the present unit-sphere treatment. The integrated route cost is also dimensionless because the unit-sphere line element ds is an angle in radians. A physical energy law enters later through a scale ϵ with units of energy per integrated phase-path-cost unit.

3. Earth-forced meridian trajectories

The constant Earth/CMB direction is projected into the tangent plane of the unit sphere. The pull response law is:

$$dn/dt = \Omega_E [\hat{c} - (\hat{c} \cdot n)n]$$

Every solution lies in a plane through the origin containing $\hat{c}$. Let $b(\theta)$ be any unit vector perpendicular to $\hat{c}$. A complete antapex-to-apex pull trajectory is:

$$n_{\theta}(s) = \sin(s)\hat{c} + \cos(s)b(\theta),$$

$$-\pi/2 \leq s \leq +\pi/2$$

At $s = -\pi/2$ the state is $-\hat{c}$, the CMB antapex. At $s = +\pi/2$ the state is $+\hat{c}$, the CMB apex. The route is one semicircle of the great circle whose plane is spanned by $\hat{c}$ and $b(\theta)$. Its plane normal is $m(\theta) = \hat{c} \times b(\theta)$, so every forced plane satisfies $m \cdot \hat{c} = 0$.

Two distinct forced planes both contain the one-dimensional line spanned by $\hat{c}$. Their intersections with the unit sphere therefore meet only at $+\hat{c}$ and $-\hat{c}$. This establishes the closure theorem: a switch between distinct Earth-forced meridians can occur only at the CMB apex or antapex. The fundamental nontrivial closed route is consequently a two-meridian spherical lune.

Two-meridian spherical lune and its route planes

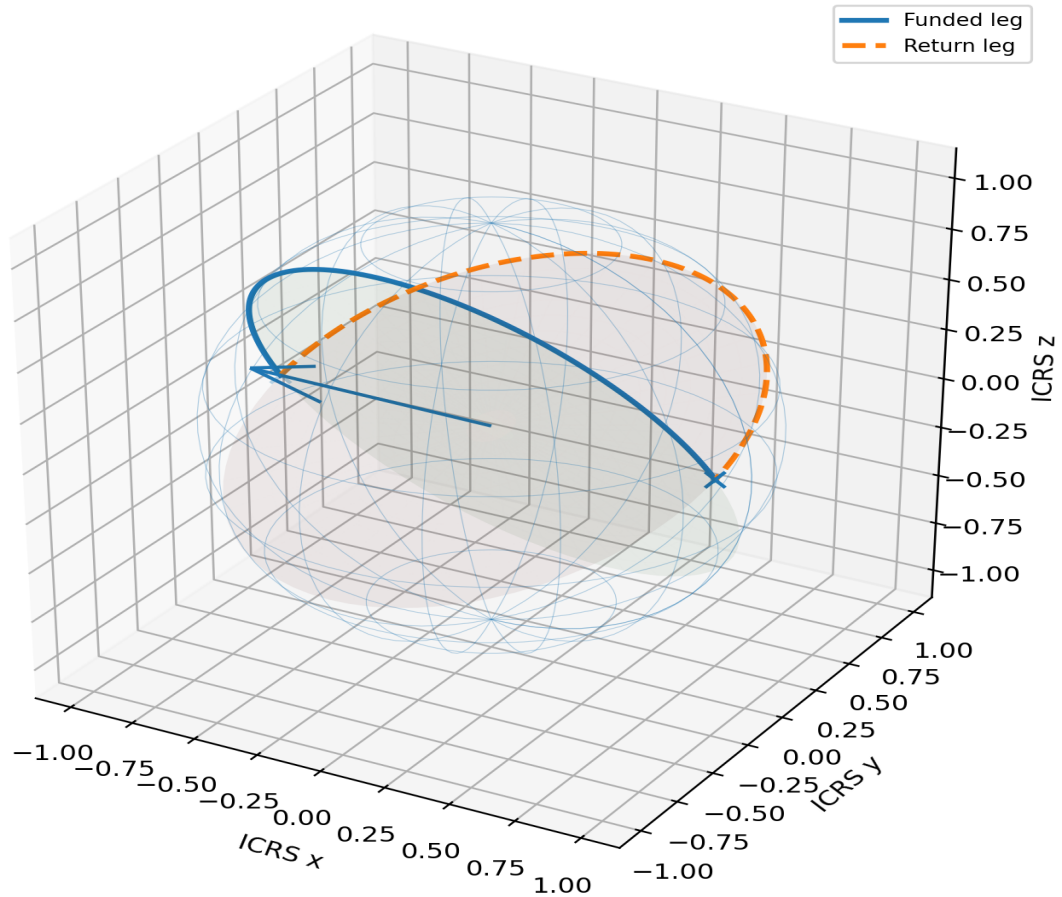


Figure 1. Three-dimensional unit sphere with the maximum-cost funded meridian, minimum-cost return meridian, their two planes, and the shared CMB endpoints. The two semicircles close exactly into a spherical lune.

4. Integrated path cost and closed-cycle objective

Peak-to-peak modulation is not the correct objective for residual energy. Residual depends on the total cost accumulated over the complete path. For a unit-sphere trajectory γ , define:

$$C[\gamma] = \int_{\gamma} f_G(n) ds$$

For the Earth meridian indexed by theta:

$$C(\theta) = \int_{-\pi/2}^{+\pi/2} \sqrt{n_{\theta}(s)^T G_{\text{phase}} n_{\theta}(s)} \, ds$$

The two semicircular branches of one plane have equal integrated cost. Replacing b by $-b$ produces the antipodally reversed path, and $f_G(-n) = f_G(n)$. Therefore the cost is a projective function of the plane normal rather than an oriented-branch function.

Let θ_E select the Earth-funded antapex-to-apex leg, and let θ_R select the apex-to-antapex return. The ideal dimensionless residual is:

$$R(\theta_E, \theta_R) = C(\theta_E) - C(\theta_R)$$

The two variables are independent because all meridians have the same endpoints. The global optimization therefore separates:

$$R_{\text{max}} = \max_{\theta} C(\theta) - \min_{\theta} C(\theta)$$

This is an exact optimization theorem for the stated route class. No visual path selection or local greedy rule is required.

5. Numerical optimization method

An orthonormal basis e_1, e_2 is constructed for the plane perpendicular to c_{hat} . The complete projective family is parameterized by $b(\theta) = \cos(\theta)e_1 + \sin(\theta)e_2$ for $0 \leq \theta < \pi$. For every θ , the route integral is evaluated directly by adaptive quadrature. A dense projective sweep locates the global maximum and minimum neighborhoods, followed by bounded scalar refinement.

The route extrema are then refined independently along each selected semicircle. All route coordinates, local costs, cumulative integrals, and Cell3 nodes are retained in the companion machine-readable file. Ten complete numerical audits verify the unit-sphere constraints, forced-plane constraints, endpoint closure, integral values, Earth-force work identities, and native Cell3 geometry.

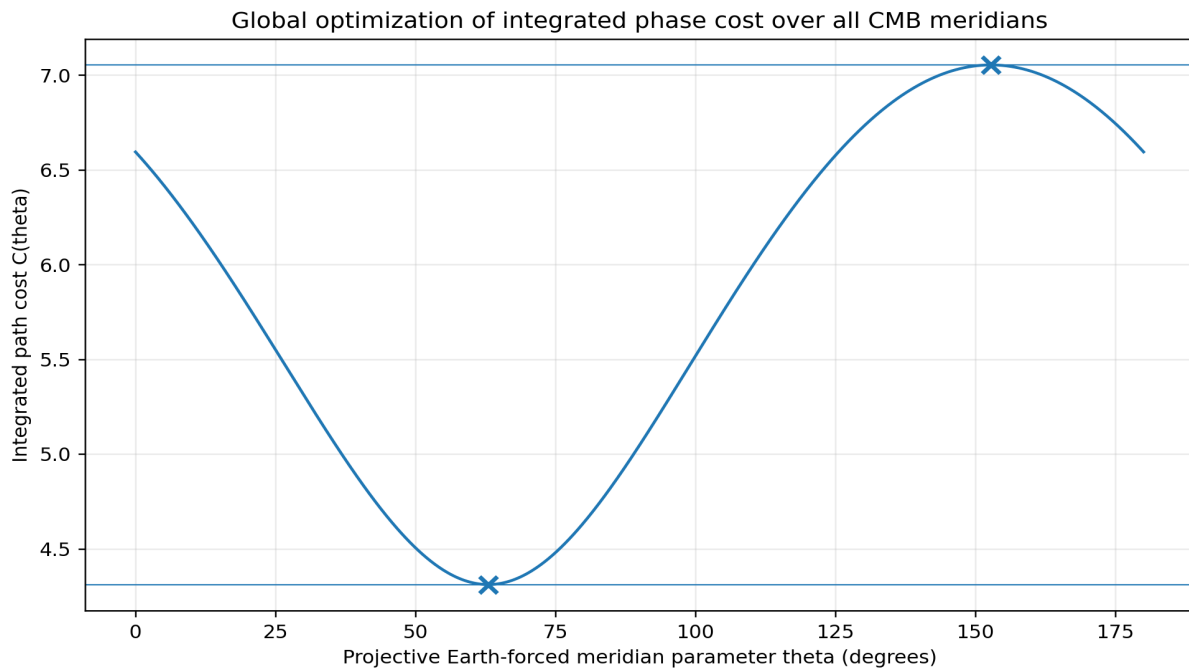


Figure 2. Integrated phase cost over the complete projective family of Earth-forced meridians. The upper marked point is the globally maximum-cost funded plane; the lower marked point is the globally minimum-cost return plane.

6. Globally maximum-cost Earth-funded leg

The optimized Earth-funded route traverses from the CMB antapex to the CMB apex. Its canonical transverse direction and plane normal are:

$$\mathbf{b}_E = (-0.200742766333689, \\ -0.425413341578701, \\ 0.882454435408164)$$

$$\mathbf{m}_E = (0.131553362649018, \\ 0.880925815222005, \\ 0.454602486631102)$$

The plane normal has RA 81.506481819880 deg and Dec +27.039360437468 deg. The antipodal normal represents the same plane. The plane equation is:

$$0.131553362649018 \ x \\ + 0.880925815222005 \ y \\ + 0.454602486631102 \ z = 0$$

The integrated path cost is $C_{E,max} = 7.053342663751261$. Dividing by the semicircle length π gives a mean route cost of 2.245148700513939.

Feature	Parameter	RA	Dec	q	f_G
CMB antapex	-90 deg	347.943299879828 deg	+6.942599922130 deg	2.142872035667380	1.463855196277070
Funded-leg minimum	-79.12427882708 2 deg	342.770949969289 deg	+16.571089216052 deg	1.897159103814636	1.377373988361416
CMB-equatorial crossing	0 deg	244.738453719460 deg	+61.939872844973 deg	8.553516543511755	2.924639557879185
Funded-leg maximum	10.875721508101 deg	224.875325379969 deg	+57.543363452211 deg	8.799229475364502	2.966349520094438
CMB apex	+90 deg	167.943299879828 deg	-6.942599922130 deg	2.142872035667380	1.463855196277070

7. Globally minimum-cost return leg

The return route traverses from the CMB apex back to the antapex. Its canonical transverse direction and plane normal are:

$$\mathbf{b}_R = (0.132360055553331, \\ 0.882630481164097, \\ 0.451047945803939)$$

$$\mathbf{m}_R = (0.200211790745414, \\ 0.421865187989889, \\ -0.884276541591357)$$

The plane normal has RA 64.611585700216 deg and Dec -62.162623331470 deg. Its plane equation is:

$$0.200211790745414 \ x \\ + 0.421865187989889 \ y \\ - 0.884276541591357 \ z = 0$$

The integrated return cost is $C_{R,min} = 4.310950793504075$. Its mean route cost is 1.372218256424204.

Feature	Parameter	RA	Dec	q	f_G
CMB apex	-90 deg	167.943299879828 deg	-6.942599922130 deg	2.142872035667380	1.463855196277070
Return-leg maximum	63.330435928413 deg	12.816990326321 deg	+18.087278077122 deg	2.310470767409517	1.520023278574877
CMB-equatorial crossing	0 deg	81.471425737100 deg	+26.810938964691 deg	1.646154044915247	1.283025348508457
Return-leg minimum	-26.66956217670 5 deg	109.798920542519 deg	+20.414357645776 deg	1.478555313173110	1.215958598461769
CMB antapex	+90 deg	347.943299879828 deg	+6.942599922130 deg	2.142872035667380	1.463855196277070

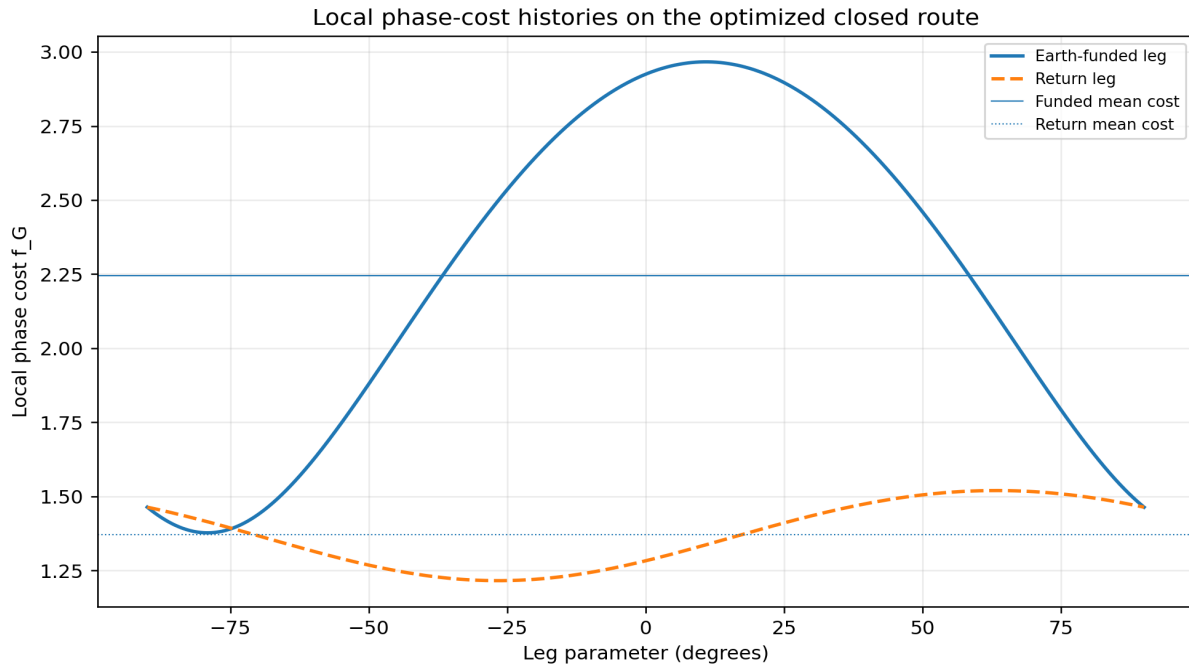


Figure 3. Local phase-cost histories of the optimized funded and return legs. The funded leg spends most of its angular length at substantially higher cost, while the return remains near the soft portion of the phase surface.

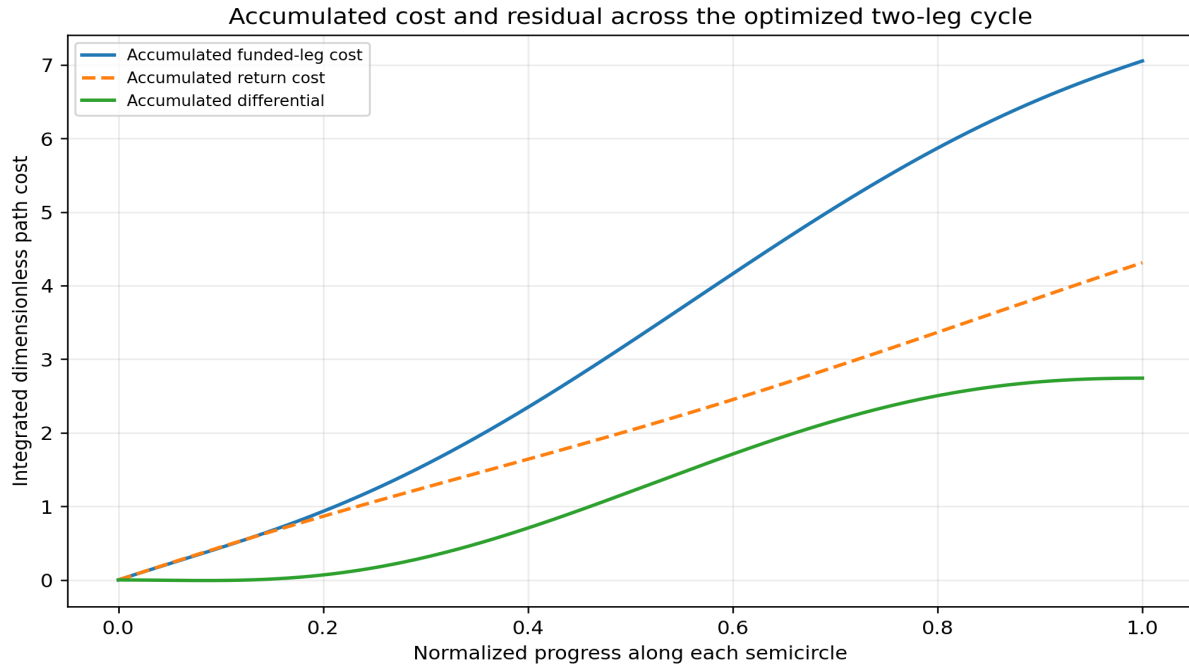


Figure 4. Cumulative integrated cost of both legs and their cumulative differential. The final differential is the ideal dimensionless residual before switching and dissipative losses.

8. Closed spherical lune geometry

The projective angle between the two optimized plane normals is 89.769450000754 deg. A pair of great-circle planes separated by angle α forms a spherical lune of solid angle 2α . The selected loop therefore encloses 3.133544940435163 sr, equal to 24.9359583335 percent of the unit sphere.

```
alpha = 1.566772470217581 rad
Omega_lune = 2 alpha = 3.133544940435163 sr
Omega_lune / (4 pi) = 0.249359583335427
```

Each optimized plane has two complementary semicircular branches. Antipodal symmetry gives equal integrated cost to the two branches of the same plane. The canonical branch pair displayed in this report is one of four equivalent branch combinations. The alternative combinations produce antipodal or complementary lunes with the same funded cost, return cost, and residual.

9. Maximum residual and residual function

The globally maximum ideal residual within the stated single-pass two-meridian route class is:

$$\begin{aligned} R_{\max} &= C_{E,\max} - C_{R,\min} \\ &= 7.053342663751261 \\ &\quad - 4.310950793504075 \\ &= 2.742391870247186 \end{aligned}$$

The minimum return consumes 61.1192593217 percent of the funded-leg integrated cost. The remaining fraction is 38.8807406783 percent. Distributed over the complete closed-loop perimeter 2π , the residual density is 0.436465222044867 path-cost units per radian.

Let C_E denote the integrated cost assigned to any available Earth-funded meridian. Once the return is fixed to the global minimum, the maximum residual is a linear function:

$$\begin{aligned} R_{\max}(C_E) &= C_E - 4.310950793504075, \\ 4.310950793504075 &\leq C_E \leq 7.053342663751261 \end{aligned}$$

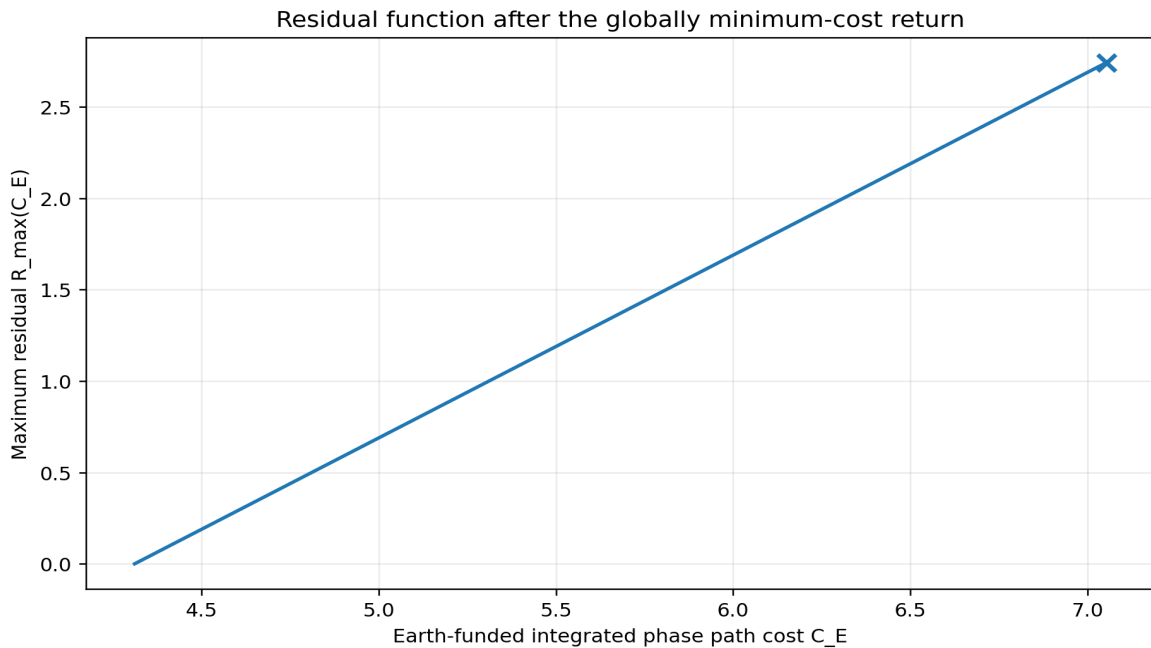


Figure 5. Maximum residual as a function of the selected Earth-funded path cost after pairing with the globally minimum-cost return. The marked endpoint is the global maximum.

Introduce epsilon as the physical energy represented by one dimensionless integrated phase-path-cost unit. The cycle energy balance is then:

$$E_{\text{res}}(C_E) = \text{epsilon} [C_E - C_{R,\min}] - E_{\text{switch}} - E_{\text{loss}}$$

At the global route optimum:

$$E_{\text{res},\max} = 2.742391870247186 \text{ epsilon} - E_{\text{switch}} - E_{\text{loss}}$$

$$E_{\text{res},\max} = 0.388807406783306 E_{E,\max} - E_{\text{switch}} - E_{\text{loss}}$$

Positive retained output requires $E_{\text{switch}} + E_{\text{loss}} < 2.742391870247186 \text{ epsilon}$.

10. Why this path differs from modulation extrema

The maximum-modulation route maximizes the difference between the highest and lowest local costs on one great circle. The present funded route maximizes the integral of cost over the entire semicircle. These are different functionals. A route can have a large pointwise range but spend little angular distance near its high-cost region; another route can have a slightly smaller range but a larger mean cost and therefore a larger integral.

The funded integral optimum is displaced by 1.815315636437 deg from the previously identified maximum-modulation plane. The minimum-integral return is displaced by 15.519957756248 deg from the least-modulated plane. These offsets are required by the change from a range objective to an integral objective.

11. Earth-force work and the required coupling asymmetry

The bare tangent-projection Earth force with magnitude coefficient F_0 is:

$$F_E(n) = F_0 [\hat{c} - (\hat{c} \cdot n)n]$$

Because $n \cdot dn = 0$ on the unit sphere, the work on any antapex-to-apex meridian is:

$$\begin{aligned} W_E &= \int F_E \cdot dn \\ &= F_0 \int \hat{c} \cdot dn \\ &= F_0 \hat{c} \cdot [\hat{c} - (-\hat{c})] \\ &= 2 F_0 \end{aligned}$$

This result is independent of the selected meridian. Direct numerical integration gives $W_E/F_0 = 2$ on the funded leg. If the same coupling remains active on the apex-to-antapex return, the return work is $-2 F_0$ and the complete closed-loop Earth-force work is zero.

The geometric residual can therefore be physically retained only when the device changes its coupling state between the two legs. The high-cost leg must be Earth-funded while the low-cost return suppresses, reverses, phase-shifts, or otherwise alters the Earth coupling. The switching mechanism and its energy cost are part of the physical device model and are explicitly included as E_{switch} .

This does not alter the route optimization. It identifies the minimum additional structure required to convert the route asymmetry into retained energy: a state-dependent coupling law with lower energetic cost than the route residual.

12. Physical interpretation of the optimized cycle

The funded leg is a high-mean-cost corridor. It begins at the common intermediate CMB endpoint cost, passes through a low point near the antapex, then climbs through the global hard-lobe neighborhood before returning to the common apex cost. The return leg occupies a different great-circle plane and remains predominantly in the low-cost region. It returns to the same antapex state with substantially less integrated phase cost.

The cycle is best understood as a route-selective loading and unloading architecture. The Earth trajectory fixes the allowed meridional directions and endpoints. The phase metric determines the cost distribution. A controllable coupling state selects which meridian is funded and which meridian is used for return. The residual is not a difference between endpoint potentials; the endpoints are identical. It is a difference between integrated route costs under asymmetric coupling.

The nearly right-angle separation of the planes creates a large spherical lune and strong route diversity. The funded route crosses the hard-axis region, while the return is aligned near the soft region. This geometric separation is the origin of the 38.8807406783 percent ideal path-cost differential.

13. Minimum mechanism requirements and testable consequences

A physical implementation sufficient to realize the optimized cycle requires four elements: a directional state n that follows or can be steered among CMB meridians; a measurable physical quantity proportional to the integrated phase cost; a switchable coupling state that distinguishes the funded and return legs; and a load channel that receives the residual after switching and dissipation.

The route model generates direct experimental predictions. Maximum loading should occur when the controlled state follows the funded plane normal at RA 81.506481819880 deg and Dec +27.039360437468 deg. Minimum return cost should occur for the return plane normal at RA 64.611585700216 deg and Dec -62.162623331470 deg. Rotating either plane away from its optimum must change the integrated response according to the one-parameter cost curve in Figure 2. Reversing to the antipodal branch should preserve the integrated cost because the phase surface is antipodally symmetric.

A complete cycle test must measure the funded-leg input, return-leg requirement, switching cost, losses, and reaction on the Earth-coupled reservoir. The central numerical target is a route-dependent differential proportional to 2.742391870247188 epsilon per ideal cycle.

14. Native Cell3 base geometry

The route computation occurs on the ICRS unit sphere. The authoritative K-Field base cell is recorded in native unit-sphere-derived lengths. Its ordered primitive translation vectors are:

$$a = (0.0203814385169464492, -0.00341641449641054938, -0.00233296591672896075)$$

$$b = (0.00219874596832335531, 0.0129021702586925406, -0.0340665015360557127)$$

$$c = (-0.000630379485347372023, -0.0242577220171278675, -0.042924354571093816)$$

Authoritative native Cell3: all 8 vertices and 27 centered half-step nodes

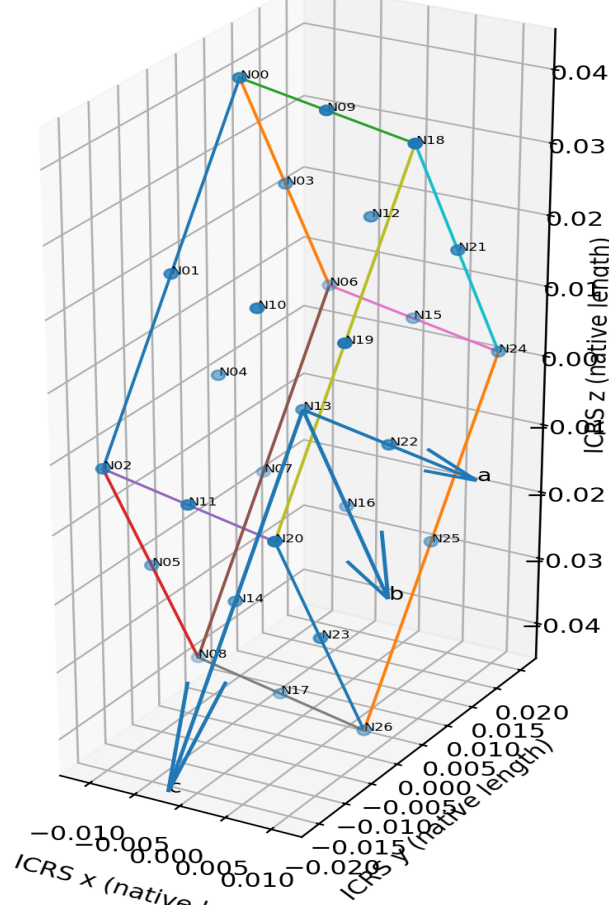


Figure 6. Dimensionally correct native Cell3 in the authoritative ICRS orientation. All eight centered vertices and all 27 centered half-step nodes are displayed. Node labels correspond to the complete numerical table below.

Cell3 primitive face planes, vertices, and oriented normals

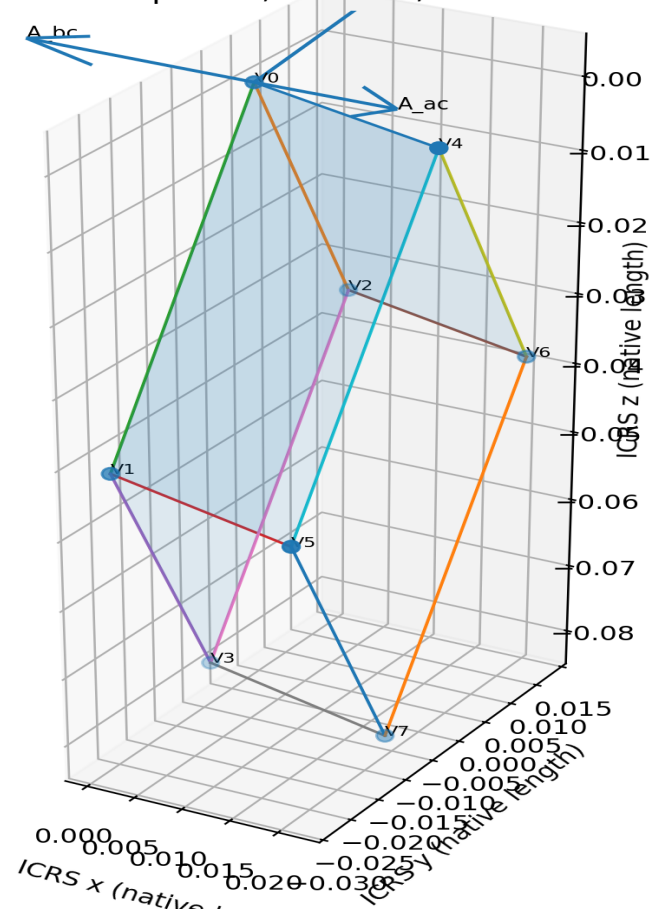


Figure 7. Primitive Cell3 parallelepiped with all eight vertices, three primitive face planes, and oriented ICRS face normals.

Vector	ICRS x	ICRS y	ICRS z	Native length
a	0.0203814385169464492	-0.00341641449641054938	-0.00233296591672896075	0.0207970587824329485
b	0.00219874596832335531	0.0129021702586925406	-0.0340665015360557127	0.0364942051307271878
c	-0.000630379485347372023	-0.0242577220171278675	-0.042924354571093816	0.0493085658998594217

Face	Area [native^2]	Included angle	Unit normal (ICRS)
A_ab	0.00075472178967262403	83.933483012313 deg	(0.194092200808, 0.913177168821, 0.358379234791)
A_ac	0.00101125575768806525	80.448130018338 deg	(0.089052594102, 0.866576769291, -0.491033948326)
A_bc	0.00138578443173328707	50.363231813756 deg	(-0.995965190309, 0.083602162507, -0.032619290564)

The primitive volume is $|\det[a \ b \ c]| = 2.84206693780964857e-05 \text{ native}^3$. The ordered basis is left-handed; the symmetric metric quantities, face areas, volumes, and route calculations remain unambiguous.

14.1 Complete centered vertex table

Vertex	Coefficient triple	ICRS x	ICRS y	ICRS z
V0	(-0.5, -0.5, -0.5)	-0.0109749024999612	0.00738598312742294	0.0396619110119392
V1	(-0.5, -0.5, 0.5)	-0.0116052819853086	-0.0168717388897049	-0.00326244355915457
V2	(-0.5, 0.5, -0.5)	-0.00877615653163786	0.0202881533861155	0.00559540947588353
V3	(-0.5, 0.5, 0.5)	-0.00940653601698523	-0.00396956863101239	-0.0373289450952103
V4	(0.5, -0.5, -0.5)	0.00940653601698523	0.00396956863101239	0.0373289450952103
V5	(0.5, -0.5, 0.5)	0.00877615653163786	-0.0202881533861155	-0.00559540947588353
V6	(0.5, 0.5, -0.5)	0.0116052819853086	0.0168717388897049	0.00326244355915457
V7	(0.5, 0.5, 0.5)	0.0109749024999612	-0.00738598312742294	-0.0396619110119392

14.2 Complete centered half-step node table

Node	Coefficient triple	ICRS x	ICRS y	ICRS z
N00	(-0.5, -0.5, -0.5)	-0.0109749024999612	0.00738598312742294	0.0396619110119392
N01	(-0.5, -0.5, 0)	-0.0112900922426349	-0.004742877881141	0.0181997337263923
N02	(-0.5, -0.5, 0.5)	-0.0116052819853086	-0.0168717388897049	-0.00326244355915457
N03	(-0.5, 0, -0.5)	-0.00987552951579954	0.0138370682567692	0.0226286602439114
N04	(-0.5, 0, 0)	-0.0101907192584732	0.00170820724820527	0.00116648295836448
N05	(-0.5, 0, 0.5)	-0.0105059090011469	-0.0104206537603587	-0.0202956943271824
N06	(-0.5, 0.5, -0.5)	-0.00877615653163786	0.0202881533861155	0.00559540947588353
N07	(-0.5, 0.5, 0)	-0.00909134627431155	0.00815929237755154	-0.0158667678096634
N08	(-0.5, 0.5, 0.5)	-0.00940653601698523	-0.00396956863101239	-0.0373289450952103
N09	(0, -0.5, -0.5)	-0.000784183241487992	0.00567777587921766	0.0384954280535748
N10	(0, -0.5, 0)	-0.00109937298416168	-0.00645108512934627	0.0170332507680279
N11	(0, -0.5, 0.5)	-0.00141456272683536	-0.0185799461379102	-0.00442892651751905
N12	(0, 0, -0.5)	0.000315189742673686	0.0121288610085639	0.0214621772855469
N13	(0, 0, 0)	0	0	0
N14	(0, 0, 0.5)	-0.000315189742673686	-0.0121288610085639	-0.0214621772855469
N15	(0, 0.5, -0.5)	0.00141456272683536	0.0185799461379102	0.00442892651751905
N16	(0, 0.5, 0)	0.00109937298416168	0.00645108512934627	-0.0170332507680279
N17	(0, 0.5, 0.5)	0.000784183241487992	-0.00567777587921766	-0.0384954280535748
N18	(0.5, -0.5, -0.5)	0.00940653601698523	0.00396956863101239	0.0373289450952103
N19	(0.5, -0.5, 0)	0.00909134627431155	-0.00815929237755154	0.0158667678096634
N20	(0.5, -0.5, 0.5)	0.00877615653163786	-0.0202881533861155	-0.00559540947588353
N21	(0.5, 0, -0.5)	0.0105059090011469	0.0104206537603587	0.0202956943271824
N22	(0.5, 0, 0)	0.0101907192584732	-0.00170820724820527	-0.00116648295836448
N23	(0.5, 0, 0.5)	0.00987552951579954	-0.0138370682567692	-0.0226286602439114
N24	(0.5, 0.5, -0.5)	0.0116052819853086	0.0168717388897049	0.00326244355915457
N25	(0.5, 0.5, 0)	0.0112900922426349	0.004742877881141	-0.0181997337263923
N26	(0.5, 0.5, 0.5)	0.0109749024999612	-0.00738598312742294	-0.0396619110119392

15. Reproduction procedure

The complete numerical result is reproduced by the following deterministic sequence:

1. Normalize the Earth/CMB vector $\mathbf{c}_{\text{hat}}$.
2. Construct an orthonormal basis $\mathbf{e}_1, \mathbf{e}_2$ perpendicular to $\mathbf{c}_{\text{hat}}$.
3. Parameterize $\mathbf{b}(\theta) = \cos(\theta)\mathbf{e}_1 + \sin(\theta)\mathbf{e}_2$.
4. Evaluate $\mathbf{n}_{\theta}(s) = \sin(s)\mathbf{c}_{\text{hat}} + \cos(s)\mathbf{b}(\theta)$.
5. Integrate $C(\theta) = \int \sqrt{\mathbf{n}^T \mathbf{G}_{\text{phase}} \mathbf{n}} \, ds$.
6. Maximize $C(\theta)$ for the funded leg.
7. Minimize $C(\theta)$ for the return leg.
8. Traverse the selected return in the reverse endpoint order.
9. Compute $R_{\text{max}} = C_{\text{E,max}} - C_{\text{R,min}}$.
10. Apply $E_{\text{res}} = \epsilon R_{\text{max}} - E_{\text{switch}} - E_{\text{loss}}$.

The companion JSON records the complete θ sweep, optimized vectors and plane normals, 1,441 samples for each route, local costs, cumulative integrals, extrema, Earth-force work checks, all Cell3 vertices and nodes, and all numerical validation flags.

16. Canonical result statement

The actual globally optimized single-pass closed Earth/CMB route is the two-meridian spherical lune formed by the funded plane normal at RA 81.506481819880 deg, Dec +27.039360437468 deg and the return plane normal at RA 64.611585700216 deg, Dec -62.162623331470 deg.

The funded leg has integrated phase cost 7.053342663751261; the return has cost 4.310950793504075; and the ideal residual is 2.742391870247186, equal to 38.8807406783 percent of the funded path cost. In physical units, $E_{\text{res,max}} = 2.742391870247186 \epsilon - E_{\text{switch}} - E_{\text{loss}}$.

The path result is exact for the defined route class. The remaining physical problem is the construction and measurement of a state-dependent Earth coupling that funds the high-cost leg and reduces or changes the coupling on the low-cost return while preserving a switching-plus-loss cost below the route residual.

Archival notice

Principal Investigator: P. Dwayne Esterline. This document and the contained K-Field formulations, numerical kernels, geometric constructions, analytical pathways, and emitted data products are designated PROPRIETARY CONFIDENTIAL TRADE SECRET material of Krytronx. Distribution and use are restricted. All Rights Reserved. Do Not Distribute.